

National University of Computer and Emerging Sciences  
Karachi Campus

Roll No

Section

Student Signature

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Attempt all the questions.

**CLO #1 & 2: Illustrate micro-architecture of x86 and RISC processors. Create basic assembly code using different type of addressing modes in x86 and RISC ISA(s) to solve simple-moderate problems.**

**Q1: Answer the following questions.**

**[Marks: 7\*2 = 14]**

- I. What is meant by a one-to-many and one-to-one relationship in context of x86 assembly. Use a coding example to support your answer.
- II. Explain the Direct, In-Direct and Indexed Modes of accessing a memory in x86 assemblies using an example instruction.
- III. Briefly explain the role of Virtual Machine(s) in execution of a high level language code.
- IV. Why Assembly language is not portable?
- V. Explain the difference(s) between Real and Protected modes of x86 processors.
- VI. What is the role of memory segmentation in x86 Intel processors regarding mismatch in register size and address bus size?
- VII. How many clock cycles are required in order to read a single value from main memory upon CPU request?

**CLO # 2: Create basic assembly code using different type of addressing modes in x86 and RISC ISAs to solve simple-moderate problems.**

**Q2: Using the following variable definitions answer the questions given below. [Marks: 5+5 = 10]**

**.data**

**var1 SBYTE -4,2,3,1**

**var2 WORD 1000h,2000h,3000h,4000h**

**var3 SWORD -16,-42**

**var4 DWORD 1,2,3,4,5**

- I. Draw a memory map and assign proper physical addresses (using a real address mode) to each byte stored in the above data segment (Assume DS= 0040h, and the starting offset is 0DEFh).
- II. What will be the hexadecimal value(s) of the destination operand after each of the following instructions are executed in sequence?
- a. `mov ah, var1`
  - b. `mov ah, byte ptr [var2+3]`
  - c. `movzx ebx, var2`
  - d. `xchg cx, [var2+4]`
  - e. `mov ax, var3`
  - f. `mov ax, [var3-2]`
  - g. `movsx, ebx, word ptr var4`
  - h. `add cx, var3[type var4]`
  - i. `mov al, [var1+6]`
  - j. `sub edx, [var4 + 12]` ; Here value of edx = 4000h

**CLO # 2: Create basic assembly code using different type of addressing modes in x86 and RISC ISA(s) to solve simple-moderate problems.**

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**Q3:** You have three different size arrays, each having 5 elements. You need to write the code to perform the following operation for each index value (X) ranging from 0 to 4, by using indirect addressing methods. **[Marks: 6]**

`arrD[X] = arrD[X] - ( arrB[X] + arrW[X] )`

The data segment has already been defined for you.

While writing code feel free to add any suitable comments to explain your steps.

```
.data arrB    BYTE    08h, A4h, A6h, B6h, 10h
      arrW WORD    2 Dup(12h), 2 Dup(0Ah),
      01h arrD    DWORD  5 Dup(FFh)
```

```
.code
;start coding from here
```